Aravind Pradeep

AI Engineer | Agentic Systems & RAG Pipelines | LLMOps & Evaluation

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PROFESSIONAL SUMMARY

AI Engineer with nearly two years at Perinet building the backend of an industrial-IoT platform: Python and Go services over MQTT streams, FastAPI, Docker, Kubernetes, and CI/CD, plus the model-benchmarking and marketing-analytics workstreams. Independently I design, ship, and maintain RAG, agent, and LLMOps systems: hybrid retrieval (BM25 + dense with reciprocal rank fusion) with RAGAs and MLflow evaluation, LangGraph and GraphRAG agent pipelines, QLoRA fine-tuning, and call-level observability. First author of arXiv:2607.02612 (DSD 2026). Nearly three years of professional software experience in total, including earlier enterprise work at Cognizant; M.Sc. in AI at BTU Cottbus, thesis phase.

PROFESSIONAL EXPERIENCE

Independent AI Engineering & M.Sc. Thesis Research

Open source (github.com/axon011) / BTU Cottbus | Cottbus, Germany | Jun 2026 – present

- First-author paper on adaptive ViT token compression presented at DSD 2026 (arXiv:2607.02612), thesis in progress; shipping the open-source RAG and agent portfolio below, including a live multi-agent German tutor.

AI Engineer

Perinet GmbH (20-person industrial-IoT company) | Cottbus, Germany | Jul 2024 – May 2026 (23 months)

- Developed Python and Go backend services connecting LLM workflows to live MQTT sensor streams, exposed as versioned async FastAPI REST APIs, containerized with Docker/Kubernetes and released to production through GitHub Actions and GitLab CI/CD pipelines.
- Led the containerization workstream for the AI Container umbrella project, covering PeriChat (Go-based chatbot with MapReduce retrieval) running on ARMv7 hardware, MessageSense (streaming MQTT anomaly detection) and PeriXplore (sensor data exploration), all three selected for production deployment; sole maintainer of several, handling profiling, QA and integration with the engine and ML teams.
- Owned the RAG evaluation and model-benchmarking workstream. Built the hybrid retrieval and evaluation layer behind PeriChat and measured retrieval speed (sub-300ms p95), answer quality and model trade-offs. The marketing team used PeriChat first, and the manager and head of product used the results to assess it for the company website and a planned pitch to Perinet's B2B clients.
- Built a marketing-analytics pipeline (CEO-sponsored) that pulls LinkedIn and Google Analytics 4 data through MQTT into a dashboard for the CEO, the head of sales and the sales and marketing teams, cutting reporting lag from roughly 48 hours to a daily refresh. Added a second dashboard where marketing enters custom events, and an event-matching script links each event to the engagement data.
- Worked with a colleague on MarketIntel, an AI competitive-intelligence tool requested by Perinet's product manager and built so he could run it without technical knowledge. Scrapling fetches competitor pages, Claude Opus extracts protocols, pricing, claims and seven KPI scores, Claude Sonnet checks each section against the page text, and a React dashboard shows side-by-side battle cards. Shipped as a Docker image built and verified in GitLab CI.
- Set up a containerized layer-2 (TAP) OpenVPN connection into Perinet's networks, giving remote access from Windows, macOS and Linux clients.
- Co-developed a voice-assistant demo that Perinet used to showcase its edge computing platform: trained a MobileNet-SSD object detection model for person detection, converted it to TensorFlow Lite and ran it on the edge device, then added a Coral USB Accelerator to improve it and extend its features; built the voice pipeline (speech-to-text, LLM, text-to-speech) and containerized and integrated the full assistant.

Software Engineer Trainee

Cognizant Technology Solutions | India | Oct 2021 – May 2022

- Developed and maintained mainframe applications (COBOL, JCL, DB2) in agile sprints; built structured debugging practices and production deployment discipline across enterprise banking systems.

PUBLICATIONS

- **Fusion: A Framework for Unified Sequential Token Adaptation in Vision Transformers.** A. Pradeep, S. Nazari, M. Taheri, C. Herglotz. Presented at DSD 2026, Krakow. arXiv:2607.02612 [cs.CV], Jul 2026.
- Stages token merging, early exiting, and token pruning so they cooperate; on ImageNet-1k with DeiT-S it matches state-of-the-art adaptive ViT methods at comparable compute while cutting inference energy 48%.

PROJECTS

Deutsch-Tutor — Adaptive AI German Tutor | tutor.aravindpradee.me | Next.js, TypeScript, SSE Streaming, Zod, Tailwind

- Designed a two-agent-per-turn flow — the reply streams instantly over SSE while a parallel Corrector returns Zod-validated exact-substring error spans — behind a provider seam that swaps the LLM backend (Claude CLI locally, Gemini when deployed) via one env var; the practice recommender is ranked by an explainable recency-weighted formula, not an LLM.
- Built and deployed the full-stack tutor (Next.js App Router, TypeScript) that logs every grammar mistake from live conversation and turns it into personalized drills: 32 CEFR-graded lessons (A1–B2), streaming chat, practice,

a progress recommender, and a 40-rule grammar reference.

RAG Evaluation System | FastAPI, Qdrant, LangChain, RAGAs, MLflow | rag-eval-system

- Engineered a hybrid retrieval pipeline (dense embeddings + BM25 + Reciprocal Rank Fusion) over a 50-topic corpus using Qdrant, reaching 0.94 hit@5 and 0.96 citation presence on a 50-question evaluation set, with async embedding pre-computation and semantic caching.
- Built an automated evaluation pipeline measuring faithfulness, relevance, and context recall via RAGAs; tracked prompt and retrieval experiments in MLflow with regression alerts when retrieval quality dropped below baseline.

GraphRAG Agent | Python, networkx, LLM Entity/Relation Extraction, Docker | graphrag-agent

- Built a GraphRAG pipeline that constructs a typed, deduplicated knowledge graph from a document corpus via LLM entity/relation extraction (networkx), so retrieval follows explicit relationships instead of chunk-embedding similarity alone.
- Implemented k-hop subgraph retrieval producing grounded answers cited back to source, handling multi-document and relationship-spanning questions that flat RAG misses; LLM backend pluggable across Claude, Codex, and Gemini with a test suite.

Multi-Agent Research & Report Pipeline | LangGraph, CrewAI, Pydantic, FastAPI, Docker | multi-agent-pipeline

- Architected a 4-agent pipeline (Planner, Researcher, Writer, Critic) using LangGraph state machines with strict role boundaries and Pydantic structured outputs, producing 2,000+ word research reports with verifiable source citations.
- Added a Critic agent that audits every cited sentence against its source snippet and triggers one bounded repair pass, plus a YAML-driven eval harness that scores citation validity, citation grounding, source diversity and routing precision over a 10-topic set; served by async FastAPI with Docker and GitHub Actions CI.

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, CrewAI, Coding Agents (Claude Code, Codex), RAG, GraphRAG, Prompt Engineering, Structured Outputs (Pydantic), Tool Use, Function Calling, MCP

LLMOps & Evaluation: Langfuse (Tracing, Evals), RAGAs (Faithfulness, Relevance, Recall), MLflow, LLM-as-Judge, A/B Prompt Evaluation, Regression Detection, Guardrails, LLM Cost Tracking

ML & NLP: Python, PyTorch, Deep Learning, Hugging Face Transformers, scikit-learn, Natural Language Processing (NLP), Generative AI (GenAI), Fine-Tuning (QLoRA, PEFT), Model Compression (Quantization, Pruning), Edge AI Optimization, Chunking, Embeddings, NER, Computer Vision, Object Detection (MobileNet-SSD), TensorFlow Lite, Speech-to-Text, Vision Transformers (ViT), Explainable ML

LLMs & APIs: OpenAI, Anthropic (Claude), Gemini, DeepSeek, Open-Source LLMs (Qwen, DeepSeek, Llama)

Backend & Infra: Go, FastAPI, Data Pipelines, Data Integration, NoSQL Databases, Data Visualization, Automated Testing, REST/gRPC, Microservices, Distributed Systems, API Integrations, Docker, Kubernetes, Container Platforms, Linux, OpenVPN, Networking (Layer 2/TAP VPN), Vector Databases, GitHub Actions, GitLab CI/CD, SQL, PostgreSQL, Qdrant, ChromaDB, OpenSearch, Elasticsearch, Azure, AWS (S3, Bedrock, EC2), Google Cloud (GCP)

Frontend: JavaScript, TypeScript, React, Next.js, Recharts

Languages: English (C1), German (B1), Malayalam (Native)

EDUCATION

M.Sc. Artificial Intelligence (Research Profile)

Brandenburg University of Technology | Cottbus, Germany | Oct 2022 – present (thesis phase)

- Thesis: Content-Aware ViT Optimization on Edge Devices (Computer Vision, Explainable ML).

B.Sc. Computer Application

BVM Holy Cross College | Kottayam, India | Jul 2018 – Mar 2021